Edit

Interest Earned Report – Functional Requirements Specification

1. Purpose

Implement an **Interest Earned Report** for the lending application.

The report must calculate how much interest has been **earned during a selected reporting period**, independent of when customers actually make their payments.

The lending business supports:

- Loan terms such as 30, 60, 90 days, or custom terms.
- Fixed interest amounts/rates per loan.
- Irregular customer payments.
- Daily, weekly, monthly, every X weeks, or unscheduled payments.
- Partial payments.
- Early settlement.
- Loan extensions with additional interest/charges.
- Manual adjustment of interest when necessary.
- Overdue and bad loans.

Because payment schedules are inconsistent, **earned interest must be based primarily on elapsed time and not payment frequency**.

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2. Core Reporting Principle

The application must distinguish between:

2.1 Contract Interest

The interest agreed upon when the loan was created.

Example:

```
Principal:          ₱10,000.00
Interest Rate:      3%
Contract Interest:  ₱300.00
Term:               60 days
```

2.2 Extension Interest

Additional interest or charges added when the borrower is granted additional time to pay.

Each extension must be treated as its own earning period.

2.3 Earned Interest

The portion of the contract and extension interest that has been earned based on elapsed time.

2.4 Interest Adjustment

Any reduction or increase to the agreed interest.

Example:

```
Original Interest:       ₱300.00
Early Payment Discount:  -₱50.00
Adjusted Interest:       ₱250.00
```

2.5 Interest Collected

The amount of interest actually received from customer payments.

This metric should only be populated if payment allocation between principal and interest is implemented.

2.6 Interest Receivable

Interest already earned but not yet collected.

```
Interest Receivable =
Final Earned Interest - Interest Collected
```

It must never be displayed as less than zero.

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3. Earned Interest Calculation

3.1 Original Loan Interest

Calculate a daily earning rate:

```
Daily Interest =
Adjusted Contract Interest / Loan Term Days
```

Example:

```
Principal:          ₱10,000
Interest:           ₱300
Term:               60 days

Daily Interest =
₱300 / 60
= ₱5.00 per day
```

3.2 Earned Interest as of a Date

```
Earned Interest =
Daily Interest × Earned Days
```

The amount must be capped at the adjusted contract interest:

```
Earned Interest =
MIN(
    Adjusted Contract Interest,
    Daily Interest × Earned Days
)
```

The system must never earn more original interest than the final interest amount agreed for that loan.

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4. Reporting Period Calculation

The report must calculate only interest attributable to the selected reporting period.

Example:

```
Report From: Aug 1, 2026
Report To:   Aug 31, 2026
```

```
Loan Date:   Aug 10, 2026
Due Date:    Oct 8, 2026
Term:        60 days
Interest:    ₱300
```

Daily interest:

```
₱300 / 60 = ₱5/day
```

Eligible days within the reporting period:

```
Aug 10 through Aug 31
```

The resulting amount is reported under:

Earned This Period

The implementation must use a consistent inclusive/exclusive date convention throughout the application so that interest is not accidentally earned for 61 days on a 60-day loan.

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5. Loan Extensions

Each loan extension must be calculated independently from the original loan interest.

Example:

```
Original Loan
Aug 10 – Oct 8
Interest: ₱300

Extension #1
Oct 9 – Nov 7
Extension Interest: ₱150
```

Extension daily interest:

```
Extension Daily Interest =
Extension Interest / Extension Term Days
```

Example:

```
₱150 / 30 days
= ₱5/day
```

The report must determine how many extension days fall within the selected reporting period and calculate:

```
Extension Earned This Period =
Extension Daily Interest
×
Eligible Extension Days
```

Each extension must be calculated separately before totals are aggregated.

Multiple extensions on the same loan must be supported.

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6. Effect of Customer Payments

Partial customer payments must **not automatically change the earned-interest calculation**.

Example:

```
Principal:        ₱10,000
Interest:         ₱300
Total Due:        ₱10,300
```

If the borrower pays ₱3,000 after 15 days:

```
Remaining Receivable =
₱10,300 - ₱3,000
= ₱7,300
```

The original contract interest remains ₱300 unless an authorized interest adjustment is made.

Payment frequency must therefore not control earned interest.

The calculation must work consistently whether the customer pays:

- Daily
- Weekly
- Every two weeks
- Monthly
- Irregularly
- Through multiple partial payments
- Through one lump-sum payment

7. Early Settlement

The application already allows the interest amount to be overridden.

If management reduces the interest because the borrower settles early, the final agreed interest must become the maximum interest that can be earned.

Example:

```
Original Interest: ₱300
Adjusted Interest: ₱180
```

Maximum original interest earned:

```
₱180
```

The report must not subsequently recognize the original ₱300.

The system should preserve both values for audit purposes:

```
Original Contract Interest
Adjusted Contract Interest
Adjustment Amount
Adjustment Reason
Adjustment Date
Adjusted By
```

---

8. Overdue Loans

If a loan reaches its due date without an approved extension, the original contract interest must stop accruing once the full contract interest has been earned.

Example:

```
60-day Loan
Contract Interest: ₱300
```

At day 60:

```
Earned Interest = ₱300
```

At day 90, without an extension:

```
Earned Interest = ₱300
```

The system must **not continue automatically charging daily interest after maturity**.

Additional interest after maturity must come from an explicitly recorded Loan Extension or other authorized charge.

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9. Bad Loans

Bad-loan classification must not erase previously earned interest.

If a loan is classified as Bad Loan, the report should retain the interest earned up to the appropriate reporting date.

However, users must be able to filter Bad Loans separately.

Loan classification values should include the application's configured classifications, such as:

```
Normal
Watch List
Bad Loan
```

---

10. Report Parameters

Create the following parameters.

Required

From Date

Start date of reporting period.

Default:

```
First day of current month
```

To Date

End date of reporting period.

Default:

```
Current Date
```

Validation:

```
From Date <= To Date
```

Optional

Customer

Searchable dropdown/autocomplete.

Default:

```
All Customers
```

Search using:

- Customer Code
- Customer Name

Parameter:

```
CustomerId: nullable
```

Loan Status

Default:

```
All Statuses
```

Possible values:

```
All
Active
Overdue
Paid
Written Off
```

Parameter:

```
LoanStatus: nullable
```

Loan Classification

Default:

```
All Classifications
```

Possible values should come from the application's loan classification configuration.

Example:

```
All
Normal
Watch List
Bad Loan
```

Parameter:

```
ClassificationId: nullable
```

Interest Type

Values:

```
All Interest
Original Loan Interest
Extension Interest
```

Default:

```
All Interest
```

Suggested parameter:

```
InterestType:
0 = All
1 = Original
2 = Extension
```

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11. Report Parameter DTO

Suggested .NET model:

```
InterestEarnedReportParameters

DateTime FromDate
DateTime ToDate
int? CustomerId
string? LoanStatus
int? ClassificationId
InterestType InterestType
int PageNumber
int PageSize
string? SortColumn
string? SortDirection
```

Defaults:

```
PageNumber = 1
PageSize = 25
SortColumn = LoanDate
SortDirection = DESC
```

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12. Report Header

Display:

Reports > Interest Earned

Title:

Interest Earned Report

Description:

> Interest earned during the selected reporting period, regardless of payment timing.

Provide:

```
Export Excel
Export PDF
```

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13. Summary Cards

After applying filters, display six summary cards.

Total Earned Interest

```
Original Interest Earned
+ Extension Interest Earned
+/- applicable adjustments
```

This is the primary KPI.

Original Interest

Total original-loan interest earned during the reporting period.

Extension Interest

Total extension interest earned during the reporting period.

Interest Adjustments

Total interest adjustments applicable to the report.

Negative adjustments must display as negative amounts.

Example:

```
-₱250.00
```

Interest Collected

Total interest actually collected during the selected period.

If payment allocation has not yet been implemented, this KPI should either be omitted or clearly marked unavailable rather than estimated from total payments.

Interest Receivable

Interest earned but not yet collected.

```
Interest Receivable =
MAX(0, Total Earned Interest - Interest Collected)
```

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14. Earned Interest Chart

Add:

Earned Interest by Month

Use a grouped column/bar chart.

Display:

```
Current Year
Previous Year
```

Example:

```
             2025        2026


Jan          ₱8,000      ₱8,500
Feb          ₱8,700      ₱9,200
Mar          ₱9,100      ₱10,300
...
```

The chart should allow the user to quickly compare lending income performance between years.

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15. Detailed Report Grid

Create the following columns.

Loan

Example:

```
LOA00009
```

Make Loan # clickable and navigate to the Loan Detail page.

Customer

Customer name.

Loan Date

Original loan/start date.

Due Date

Current applicable due date.

Principal

Original loan principal.

Contract Interest

Original agreed interest amount.

Extension Interest

Total extension interest applicable to the loan.

Earned Before Period

Interest already earned before `FromDate`.

This allows users to understand how much interest had already been recognized before the current report.

Earned This Period

Interest earned between:

```
FromDate and ToDate
```

This is the most important detail column.

Total Earned

Interest earned from the beginning of the loan through `ToDate`.

Adjustment

Interest adjustments applicable to the loan.

Example:

```
-₱50.00
```

Final Earned

Final recognized interest after adjustments.

Status

Examples:

```
Active
Overdue
Paid
Written Off
```

Display using existing status badges.

Classification

Examples:

```
Normal
Watch List
Bad Loan
```

Display using existing classification badges.

---

16. Grid Footer

Display totals for all relevant monetary columns:

```
Principal
Contract Interest
Extension Interest
Earned Before Period
Earned This Period
Total Earned
Adjustment
Final Earned
```

Totals must be calculated against **all records matching the report parameters**, not only the current pagination page.

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17. Pagination

Default:

```
25 rows
```

Available options:

```
25
50
100
```

Display:

```
Showing 1–25 of 84 loans
```

Server-side pagination is recommended.

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18. Loan Interest Drill-Down

Clicking the Loan # may open the existing Loan Detail page.

Optionally provide an **Interest Calculation Breakdown** section.

Example:

```
LOA00009
Ana Villanueva

ORIGINAL LOAN
Loan Period:        Aug 10 – Oct 8
Contract Interest: ₱300.00
Term:               60 days
Daily Interest:     ₱5.00
Earned Days:        22
Earned This Period:₱110.00

EXTENSION #1
Extension Period:  Oct 9 – Nov 7
Extension Interest:₱150.00
Term:               30 days
Daily Interest:     ₱5.00
Earned Days:        0
Earned This Period:₱0.00
```

This is highly recommended because users must be able to understand how the system arrived at an earned-interest figure.

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19. Suggested API

Report

```
GET /api/reports/interest-earned
```

Parameters:

```
fromDate
toDate
customerId
loanStatus
classificationId
interestType
pageNumber
pageSize
sortColumn
sortDirection
```

Example conceptual request:

```
/api/reports/interest-earned
?fromDate=2026-08-01
&toDate=2026-08-31
&customerId=
&loanStatus=
&classificationId=
&interestType=0
&pageNumber=1
&pageSize=25
```

---

20. Suggested API Response

```
{
  "parameters": {
    "fromDate": "2026-08-01",
    "toDate": "2026-08-31"
  },
  "summary": {
```

```
    "totalEarnedInterest": 12450.00,
    "originalInterestEarned": 10800.00,
    "extensionInterestEarned": 1900.00,
    "interestAdjustments": -250.00,
    "interestCollected": 9850.00,
    "interestReceivable": 2600.00
  },
  "records": [],
  "pagination": {
    "pageNumber": 1,
    "pageSize": 25,
    "totalRecords": 84,
    "totalPages": 4
  }
}
```


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21. Suggested Calculation Service

Do not place the earned-interest calculation directly inside the Angular component or API controller.

Create a dedicated service, for example:

```
IInterestCalculationService
InterestCalculationService
```

Suggested responsibilities:

```
CalculateOriginalInterestEarned()
CalculateExtensionInterestEarned()
CalculateInterestEarnedForPeriod()
CalculateInterestEarnedBeforePeriod()
CalculateTotalInterestEarned()
CalculateInterestAdjustment()
```

The Reporting service should use this calculation service.

This allows the same calculation rules to later be reused by:

- Dashboard
- Loan Detail
- Statement of Account
- Interest Earned Report

- Profitability Reports
- Year-end reports

---

22. Calculation Precision

All financial calculations must use:

```
decimal
```

in .NET.

Do not use:

```
float
double
```

for monetary calculations.

Intermediate daily-interest calculations should retain sufficient decimal precision.

Do not round the daily interest before multiplying it by the number of earned days.

Preferred:

```
Earned =
Round(
    InterestAmount / TotalDays * EarnedDays,
    2
)
```

Rather than:

```
Daily =
Round(InterestAmount / TotalDays, 2)

Earned =
Daily * EarnedDays
```

This prevents cumulative rounding differences.

---

23. Final-Day Reconciliation

Because daily allocation can create fractions of a centavo, the system must guarantee:

```
Total Interest Earned at maturity
=
Final Adjusted Contract Interest
```

Example:

```
Contract Interest = ₱100
Term = 30 days

₱100 / 30 = ₱3.333333...
```

At maturity, the final amount must be exactly:

```
₱100.00
```

and not:

```
₱99.90
₱100.01
```

The same rule applies to extension interest.

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24. Important Date Rules

Centralize the definition of a **loan term day**.

For example, if:

```
Loan Date = Aug 10
Term = 60 days
```

the application must consistently determine which 60 calendar days earn interest.

Do not independently calculate term days using different date-difference logic in the UI, reports, and backend.

Prefer storing or deriving a single authoritative:

```
TermDays
```

for each loan and extension.

25. Performance Requirements

All calculations must be performed on the server.

Angular should only:

- Send report parameters.
- Display results.
- Display charts.
- Handle pagination/sorting/filter controls.
- Trigger exports.

For larger datasets:

- Use server-side filtering.
- Use server-side sorting.
- Use server-side pagination.
- Avoid loading every payment record when only loan-level calculations are required.
- Retrieve extension records efficiently.
- Avoid N+1 database queries.

---

26. PDF Export

The PDF version should contain:

```
Business Name
Interest Earned Report
Reporting Period
Selected Filters
Generated Date/Time

Summary

Detailed Loan Interest

Totals
```

The PDF should use the same calculations as the screen report.

Do **not** recalculate values independently in the PDF-generation code.

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27. Excel Export

Excel should include at minimum:

```
Loan #
Customer
Loan Date
Due Date
Principal
Contract Interest
Extension Interest
Earned Before Period
Earned This Period
Total Earned
Adjustment
Final Earned
Status
Classification
```

Use numeric Excel cells for monetary values rather than exporting formatted currency strings.

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28. Auditability

The report must be reproducible.

Any manual change affecting interest should retain:

```
Original Value
New Value
Reason
Changed By
Changed Date/Time
```

This applies especially to:

- Contract interest changes
- Interest discounts
- Extension interest

- Extension dates
- Loan dates
- Due dates
- Bad-loan classification
- Write-offs

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29. Important Business Rule

Payments and earned interest are separate concepts.

A payment reduces the customer's outstanding receivable.

A payment does not determine how much interest has been earned.

```
Outstanding Receivable =
Principal
+ Contract Interest
+ Extension Interest
- Payments
```

while:

```
Earned Interest =
Interest attributable to elapsed eligible days
```

This separation must be maintained throughout the implementation.

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30. Future Reports

Design the reporting architecture so the same infrastructure can later support:

1. Collections Report
2. Loan Releases Report
3. Outstanding Receivables Report
4. Overdue Loans Report
5. Bad Loan Report
6. Cash Flow Report
7. Customer Loan History Report
8. Loan Profitability Report
9. Monthly/Annual Lending Performance Report

The Interest Earned Report should be implemented as the first report using reusable reporting components and calculation services.